

Full-Chain Multi-Hop Retrieval with Reified Assertion Graphs, Canonical Entity Memories, and Bounded Evidence Planning

Evaluation under strict model parity on 2WikiMultihopQA and HotpotQA

September 2026

FC@5: **97.7** vs. 65.8 (HippoRAG 2) / 67.6 (CatRAG) on 2Wiki ð **93.0** vs. 80.4 on HotpotQA, zero tuning ð ~\$0.001/query

Roadmap

1. **The problem:** chains break while recall looks fine — and its two structural causes.
2. **The representation:** reified assertions over canonical entities; an audited memory; a layered graph with a closed-form walk.
3. **The planner:** three LLM roles under contracts — analyst, repair hop, set packer.
4. **The evidence:** parity protocol, main results, depth analysis, ablations, a no-graph control, and the system's evolution.

Every number in this talk is a single pre-registered measurement, paired per question with bootstrap CIs; every LLM call is cached (zero-cost reproduction).

The problem: high recall, broken chains

The task metric must be the **complete chain**:

$$R@k = \frac{|G_q \cap T_k(q)|}{|G_q|}, \quad FC@k = \mathbf{1}[G_q \subseteq T_k(q)]$$

A reader cannot compose an answer from 3 of 4 passages: $R@5 = 0.75$ is *operationally zero*.

double-bridge stratum (2Wiki)	R@5	FC@5
BM25	50.9	0.9
HippoRAG 2 (its code, our run)	—	12.3



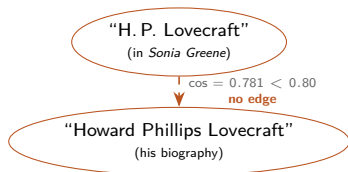
the chain
breaks here

top-5 returned:
A, dir.A, B, 2 distractors

The signature of the state of the art: *partial* recall stays high while the set misses exactly the bridge passage.

Two structural causes, both measured

1 û Identity unresolved at indexing.



Thresholded synonymy (172k edges in HippoRAG 2) fails where it is most needed: **57% of chain failures were identity islands.**

2 û Resources fixed per query.

- one ≤ 4 -triple selection budget,
- one seed distribution,
- one slack position in the top-5,

shared across all chains of the question. A double chain is starved **by construction**: covering one branch guarantees $FC = 0$ regardless of ranking quality.

Both causes are properties of the *design*, not of model quality — so both are removable by design.

Contributions

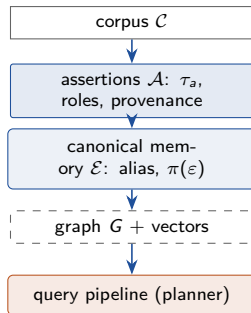
1. **Reified-assertion graph over canonical entities** — facts as first-class, query-modulable nodes; identity resolved *before* the graph; no synonymy edges.
2. **Audited entity memory** — merge decisions with provenance, audited against external Wikidata gold: over-merges $74 \rightarrow 32$.
3. **Bounded evidence planner** — analyst / repair / packer under JSON contracts; ≤ 3 LLM calls; deterministic control flow.
4. **Structural analysis** — the layered walk has a closed form: *the walk cannot chain hops; the seeding does*.
5. **Evidence** — strict parity, paired statistics, ablations, a no-graph control, a depth (accumulation-vs-collapse) analysis, and frozen-configuration generalization.

Architecture and notation: pay once at indexing, decide well at query time

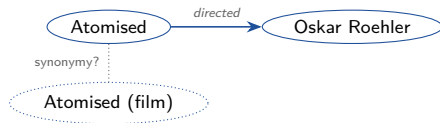
Indexing (once per corpus, no questions): passages $\mathcal{C} \rightarrow$ assertions $\mathcal{A}$ (extraction) $\rightarrow$ entities $\mathcal{E}$ (canonical memory) $\rightarrow$ graph $G = (\mathcal{C} \cup \mathcal{E} \cup \mathcal{A}, E) +$ aligned vector stores $e(\tau_a)$, $\phi(\varepsilon)$, $e(c)$.

Query (8 steps, ≤ 3 LLM calls): affinity σ ; linking $L(q)$; analyst (plan, $S(q)$, slots); seeding $\mathbf{v}$; modulation μ ; walk $\mathbf{p}^*$; conditional repair; set packing.

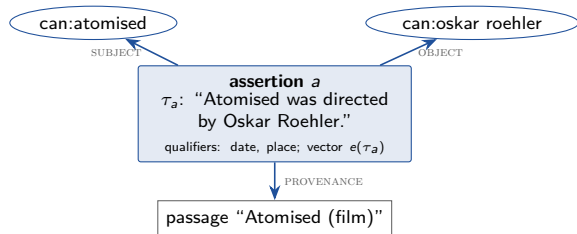
Real sizes (2Wiki): 6,119 passages $\rightarrow$ 47,999 assertions, 27,966 entities, 165k edges; $\sim \$3$, once.



Representation I: the fact as a node, not an edge



(a) HippoRAG 2: the fact is a binary, anonymous edge, no provenance

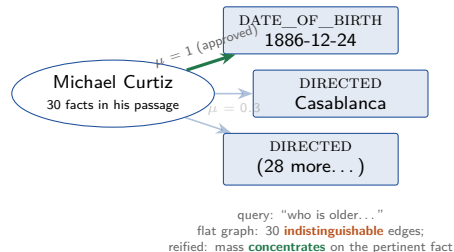


(b) ours: a node with its own sentence, n -ary roles and embedding, on canonical entities

- We embed **the sentence**, not the label: robust to extractor noise; the datum the question compares (dates, doses) travels *inside* the retrievable unit.
- The walk reasons **through the fact** and its transit is per-query: $\mu_a = \frac{1}{4} + \frac{3}{4} \tilde{\sigma}_a(q)$.

Representation II: why the assertion-node matters (three properties)

1. **Query modularity.** A triple graph fixes the transition distribution at indexing; CatRAG re-weights the same binary support. Here each fact is *addressable*:
 $P(\varepsilon \rightarrow a) \propto \mu_a(q)$, approved facts transit at full weight.
2. **n -ary closure.** Triples force binarity; extra participants and qualifiers are lost. The assertion keeps them in τ_a and its roles.
3. **Sentence semantics.** Affinity on $e(\tau_a)$, a self-contained sentence — not a concatenation of noisy labels.



Canonical entity memory: kill the islands in the index

canonical entry can:h. p. lovecraft
alias: {H. P. Lovecraft;
Howard Phillips Lovecraft}
description: "American
writer of weird fiction. . ."
own passage $\pi(\varepsilon)$: *H. P. Lovecraft*

Mention cards (name + context) $\rightarrow$ candidates on 2
channels $\rightarrow$ hard constraints (dates, ordinals,
disambiguators) $\rightarrow$ LLM adjudication (strict:
relatives, successors, remakes are *not* the same) $\rightarrow$
transitive closure. Built **once, from the corpus only**.

Per-pair decision (name channel ν , context
channel ϕ , hard constraints H):

$$d(m, m') = \begin{cases} \text{VETO} & \neg H \\ \text{MERGE} & \text{fold}(\nu_m) = \text{fold}(\nu_{m'}) \\ \text{MERGE} & \langle \nu, \nu' \rangle \geq 0.90 \wedge \langle \phi, \phi' \rangle \geq 0.70 \\ \text{JUDGE} & \langle \nu, \nu' \rangle \geq 0.80 \vee (\langle \phi, \phi' \rangle \geq 0.75) \\ \text{DROP} & \text{otherwise} \end{cases}$$

40,276 mentions $\rightarrow$ 27,966 entries; 178/188 diagnosed
islands resolved; the canonical graph needs **no
synonymy edges at all**.

The memory is audited against external gold (Wikidata QIDs)

2Wiki carries QIDs for its evidence entities (never used to build). Unit: (surface form, QID) of chain entities.

- entry with 2 QIDs of distinct forms = **over-merge**;
- QID split across entries = **under-merge**;
- every union carries **provenance** $\Rightarrow$ each over-merge is attributed to the rule that bridged it.

memory	over	under	dominant bridge
initial	74/2,211	6	title $\rightarrow$ subject union
curated	32	6	residual judge errors

Four registered decisions (no silent patches):

1. title $\rightarrow$ subject unions with a genuine title mention $\rightarrow$ judge (rejected 893/1,549);
2. near-identical names with differing parenthetical disambiguators $\rightarrow$ judge ("X (1922 film)" $\neq$ "X (1923 film)");
3. title cards inherit the article subject's dates;
4. provenance stored per union.

Honesty note: two dev questions relied on **fortunate wrong merges** (a dataset-mislabeled homonym; an emperor merged with his son). We give them up: identity precision is the foundation.

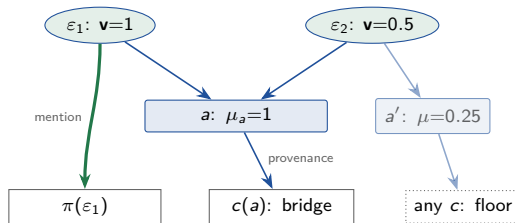
The search graph is layered — and its walk has a closed form

Proposition. Entities have in-degree 0; each assertion drains into its unique source passage; passages are sinks. Hence

$$\mathbf{p}^*(c) = \beta \left[\mathbf{v}(c) + \lambda \sum_{\varepsilon: c \in N(\varepsilon)} \frac{\mathbf{v}(\varepsilon)}{Z_\varepsilon} + \lambda^2 \sum_{\varepsilon} \sum_{\substack{a \ni \varepsilon \\ c(a)=c}} \frac{\mathbf{v}(\varepsilon)\mu_a}{Z_\varepsilon} \right]$$

verified numerically to the 4th decimal.

Consequence: the walk *diffuses* the seeds one fact away; it **cannot chain hops**. Multi-hop capability must live in **which entities and own passages get seeded** — the planner's job.

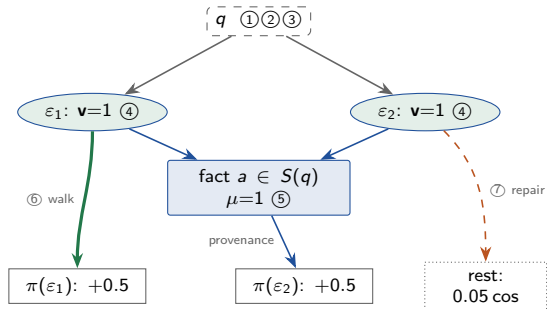


three channels only: own seed $\hat{u}$ mentions of seeded entities $\hat{u}$ modulated facts of seeded entities (depth ≤ 2)

Iterative solver kept only for cyclic variants (synonymy; clinical IS-A layers).

The query pipeline: Algorithm 1 on the graph

1. $\sigma_a = \langle e(q), e(\tau_a) \rangle$ (one matrix product).
2. Link mentions $\rightarrow L(q) \subseteq \mathcal{E}$.
3. **Analyst** on $\text{Top}_{40} \sigma \cup \text{frontier}(L)$: plan, selection $S(q)$ ($\tilde{\sigma}_5=1$), unfilled slots. LLM 1
4. Seed $\mathbf{v}$ (Eq. next slide).
5. Modulate $\tilde{A} = A \text{diag}(\mu)$; $P = D^{-1} \tilde{A}$.
6. Walk to $\mathbf{p}^*$ ($\lambda=0.5$, $\|\Delta\|_1 < 10^{-9}$).
7. Slots left $\rightarrow$ **repair hop**, re-seed, re-walk (≤ 3 rounds). LLM 2
8. **Pack** the 5-set that covers the plan. LLM 3



⑧ $\mathbf{p}^*$ restricted to passages $\rightarrow$ the set that covers the plan

The analyst: plan the chain *before* selecting

"Which film has the director died first, Bílá Spona or A Night at Earl Carroll's?"

candidate facts (top-40 cosine $\cup$ frontier)
2: A Night at Earl Carroll's ... **directed by Kurt Neumann**
6: Bílá spona was ... **directed by Martin Frič**
8: *Earl Carroll* (the actor) *died on June 17, 1948...*

plan (typed slots):

director of Bílá spona $\rightarrow$ Martin Frič

date of death of Martin Frič

director of A Night. . . $\rightarrow$ Kurt Neumann

date of death of Kurt Neumann

Without a plan (direct selector, fixed budget 4):
picked the actor Earl Carroll as "director" and omitted Kurt Neumann *who was on the list* $\Rightarrow$ broken chain.

With a plan (budget = number of slots):
selects both bridge facts; hard rule: slots may name an entity *only if a candidate fact states it* — else [X]; parametric knowledge forbidden.

The same call performs **entity recognition** (mentions $\rightarrow$ canonical form $\rightarrow$ memory) and declares the **unfilled slots** that feed the repair hop.

The analyst for real: raw input/output on a benchmark question

Input (question + 43 numbered facts = $\text{Top}_{40} \sigma \cup 3$ from the frontier):

Question: Do Walfredo Reyes Jr. and Zac Dysert have the same nationality

0: Walfredo Reyes Jr. was born on December 18, 1955. ... 17: Zac Dysert is an American football quarterback. ...

Raw output (JSON mode, temperature 0, cached):

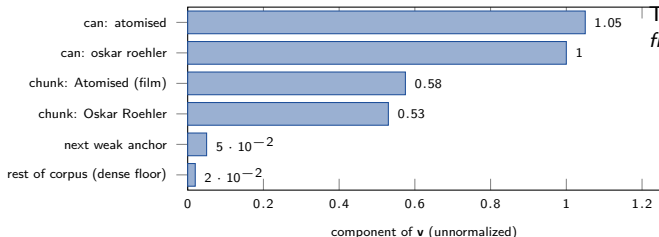
```
{"mentions": [{"text": "Walfredo Reyes Jr.", "canonical": "Walfredo Reyes Jr."}, ...],  
"plan": ["nationality of Walfredo Reyes Jr.", "nationality of Zac Dysert"],  
"selected": [17, 18], "unfilled": [{"query": "nationality of Walfredo Reyes Jr.", "anchor":  
"Walfredo Reyes Jr."}]}
```

Deterministic post-processing (the LLM only writes text and copies line numbers):

- selected are *local* indices $\rightarrow$ guarded map to global assertion ids ($17 \rightarrow \text{Zac Dysert}::a1$); dedup; dynamic cap $\min(8, \max(2, |\text{plan}|))$;
- canonical forms $\rightarrow$ linker $\rightarrow$ canonical entries (extends $L(q)$); anchor $\rightarrow$ entry $\rightarrow$ own passage $\rightarrow$ its assertions (repair candidates);
- invalid indices / malformed JSON degrade gracefully (guards; dense fallback) — thousands of calls per benchmark, zero hard failures.

Seeding: four sources, one equation, observed values

$$\mathbf{v}(u) = \underbrace{\mathbb{1}[u \in \mathcal{E}_S] \tilde{\sigma}(u)}_{\text{hard gate}} + \underbrace{\mathbb{1}[u \in \mathcal{C}] w_{\text{pas}} \hat{\mathbf{s}}(u)}_{\text{dense floor}} + \underbrace{\epsilon \mathbb{1}[u \in \text{Top}_2\langle e(q), \phi \rangle]}_{\text{weak anchors}} + \underbrace{\mathbb{1}[u = \pi(\varepsilon)] w_{\text{link}} \mathbf{v}(\varepsilon)}_{\text{own passage}}$$



Trace: “Who is the mother of the director of film Atomised?”

- can:oskar roehler = 1.0: entity of the approved fact — **discovered**, not named in the question;
- both *own passages* get $0.5 \cdot \mathbf{v}(\varepsilon)$;
- dense floor: 94–98% of raw mass, but $\sim 1.6 \cdot 10^{-4}$ per passage after normalization; a gated entity concentrates 25–60 $\times$ that.

After the walk: $p^*(\text{Atomised}) = 0.0084$,
 $p^*(\text{Roehler}) = 0.0063$, first distractor 0.0004
 — an order of magnitude.

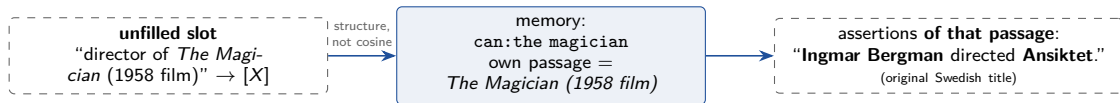
Modulated personalized PageRank

$$\tilde{A} = A \operatorname{diag}(\mu), \quad \mu_u = \begin{cases} \frac{1}{4} + \frac{3}{4} \operatorname{clip}_{[0,1]}(\tilde{\sigma}_u) & u \in \mathcal{A} \\ 1 & \text{else} \end{cases} \quad P = D^{-1} \tilde{A}$$

$$\mathbf{p}^{(t+1)} = \lambda(P^\top \mathbf{p}^{(t)} + m_{\text{dang}} \mathbf{v}) + (1 - \lambda) \mathbf{v}, \quad \lambda = 0.5$$

- Columns entering *assertion* nodes are rescaled: crossing a fact costs according to *this* question; approved facts transit at $\mu = 1$, irrelevant ones at the 0.25 floor (never blocked).
- Sinks (passages) recycle their mass to the seeds (m_{dang}): a page “votes” and restarts.
- Dials ($\lambda=0.5$, $w_{\text{pas}}=0.05$, all-passage seeding) are HippoRAG 2’s published values — **pre-registered parity**, frozen across datasets.
- Sparse products over 165k edges: milliseconds; no LLM.

The repair hop: structural candidates beat cosine where cosine is blind



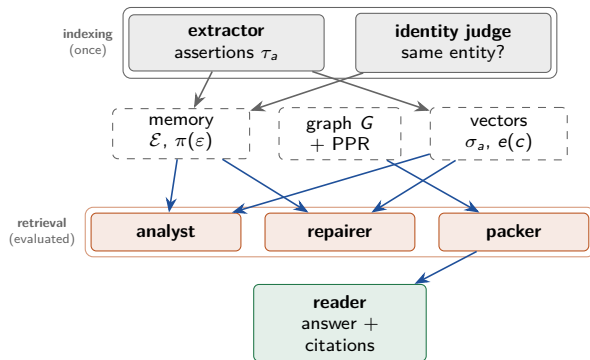
the slot's cosine with "Ingmar Bergman directed Ansiktet" is **low** (zero shared vocabulary); the **structural route** (linked entity → own passage → its facts, provenance visible) rescues it: the mini-filter approves, Bergman is seeded, his biography enters the top-2

- Runs only if the plan left slots (~15–25% of queries); up to 3 rounds with re-planning when [X] placeholders remain.
- The genuinely new capability: bridges **immune to aliasing** (languages, original titles) — unreachable for similarity and synonymy.

Set packing, and the census of LLM roles

Set packing (LLM call 3): FC@ k is a property of the set. From the walk's top-12, pick the 5 that jointly cover the plan (walk's top-2 preserved). Worth +1.5 FC@5 and +6 FC@2 on the probe; +4.0 R@2 on HotpotQA.

Bounded agency: JSON contracts validated; ≤ 3 calls/query; read-only tools; cache by (model, prompt); deterministic control flow — *the LLM decides within contracts; the algorithm keeps control.*



Protocol: strict parity, data separation, pre-registration

Parity. Same backbone everywhere (gpt-4o-mini+te3-small); HippoRAG 2 run *with its official code* on the same corpus; CatRAG compared on its published numbers under the identical backbone. Our HippoRAG 2 reproduction matches the CatRAG paper's independent reproduction **to the decimal** (85.9/65.8 vs 85.9/66.1).

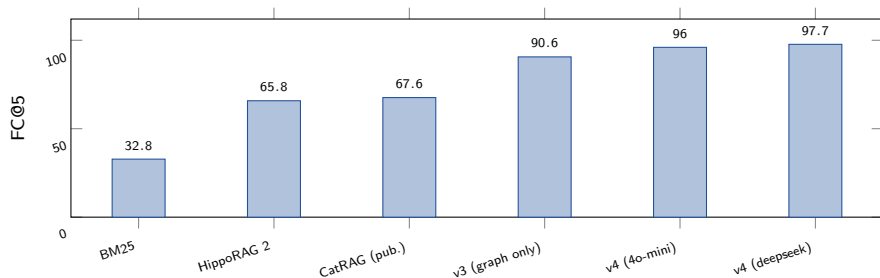
Datasets: 2WikiMultihopQA (1,000 q $\times$ 6,119 passages; 4 types, double-bridge has 4 golds) $\hat{u}$ HotpotQA (1,000 $\times$ 9,811; bridge/comparison).

Separation. All design on a 200-question held-out probe + calibration pool; test sets touched *once per configuration*; HotpotQA with **zero** dataset tuning.

Statistics. Paired per question, bootstrap $B=2,000$; significance = CI excludes 0.

Cost. Whole study $\sim \$40$; $\sim \$0.001$ /query; all calls cached $\Rightarrow$ zero-cost reproduction.

Main result (2WikiMultihopQA, all systems on the same backbone)

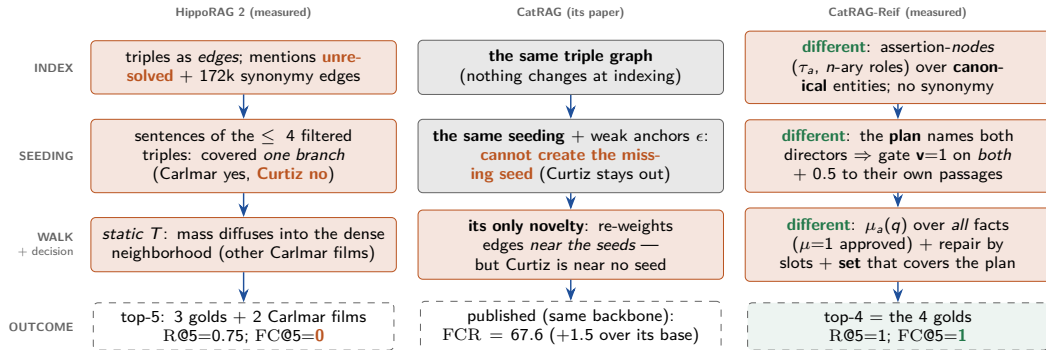


	R@2	R@5	FC@5	FC@10
HippoRAG 2 (its code)	70.7	85.9	65.8	71.7
CATRAG-REIF v4 (deepseek)	84.2	98.9	97.7	97.8

Paired $\Delta\text{FC@5} = +31.9$ [+28.9, +34.9]: wins 326 / loses 7; all four types significant. HippoRAG 2's own paper with a 7B GPU encoder reports $\text{R@5} = 90.4$; here 98.9 at a fraction of the cost.

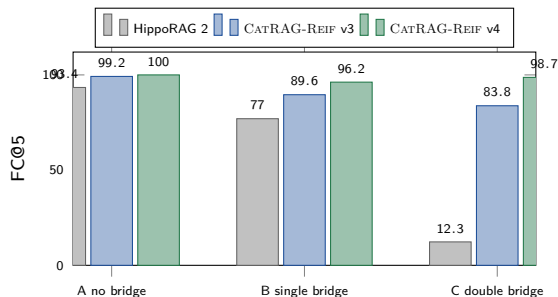
Case study: three mechanisms, stage by stage

“Which film has the director who is older, *God’s Gift to Women* or *Aldri annet enn bråk?*” — gold: 2 films + their directors (M. Curtiz, E. Carlmar). inherited from HippoRAG 2 unchanged what each system changes



Column reading: CatRAG changes *one* stage and inherits index and seeding (+1.5); CATRAG-REIF changes all four.

The central finding: accumulation versus collapse with depth



Null model: independent links

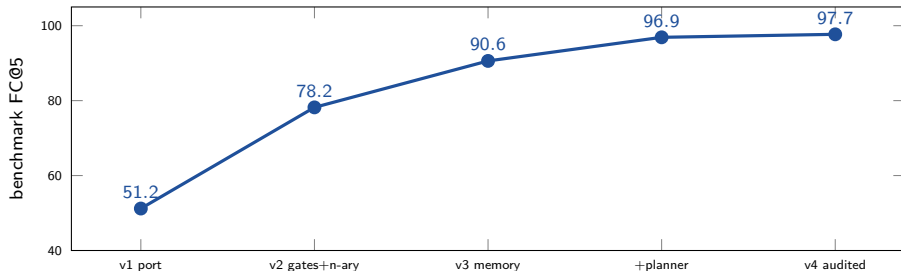
$$\Rightarrow FC \approx p^{|G_q|}$$

	C pred.	C obs.	ratio
HippoRAG 2	59.2	12.3	0.21
CATRAG-REIF v3	80.3	83.8	1.04

v3 degrades by *accumulation* ($\sim 5\%$ per link).

HippoRAG 2 *collapses*: fixed per-query resources shared across chains. The planner **inverts the gradient**: with v4 the double chain becomes the *best* stratum (98.7).

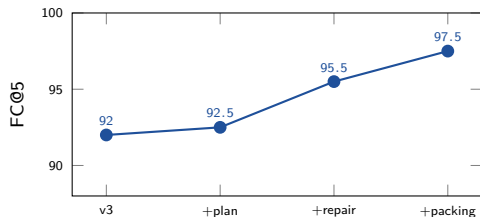
The system's evolution: design by diagnosis



step	measured diagnosis of the predecessor	remedy
→v2	filter used parametric knowledge; n -ary participants unreachable	few-shot filter, hard gate, n -ary roles
→v3	57% of failures were identity islands	canonical memory + own-passage seeding
→plan	filter <i>saw</i> the bridge but did not select it (12/15)	analyst plan, repair hop, set packing
→v4	QID audit: 74 over-merged entries; linker blind to variants	audited curation; name-to-name linker

+46 FC@5 in four iterations — every decision on held-out probes, one benchmark measurement per configuration.

Attribution I: component ablations (held-out probe)



The plan alone barely moves the aggregate (+0.5): its value is to *enable*

- the **repair hop** (+3.0; double chains 86 → 96) and
- the **set packing** (+1.5; +6 FC@2 early ordering).

Transfers to HotpotQA's probe: repair +2.0 [+0.5, +4.0], packing +4.0 R@2 [+1.5, +6.8].

Planner-model sweep (probe FC@5): deepseek 97.5 ∩ gpt-oss-20B 97.0 ∩ gpt-4o 96.5 ∩ 4o-mini 96.0 — **the design, not the scale, carries the gain**; a weak model (qwen-27B, 67.5) marks the capability floor.

Attribution II: graph or planner? A no-graph control at identical budget

Same analyst, repair, packer, prompts, demos, budget — over dense top-20 passages + a title index; **no** assertions, memory, graph, walk.

benchmark	v4	control	Δ [CI 95%]
2Wiki	96.0	83.3	+12.7 [+10.5, +14.9]
HotpotQA	90.2	83.9	+6.3 [+4.1, +8.6]

Pre-registered criterion (≥ 5 , significant on both): **met**. A post-hoc strengthened control still loses 9.6/5.1.

Where the control fails: of the golds it misses (v4 finds), 130/141 are **outside its top-20** — failure of *reach*, not of ranking. The same planner reads "... directed by Ben Holmes" in prose and writes [X]; as an atomic assertion it approves it and the memory turns approval into retrieval.

Decomposition of the margin over CatRAG:
planner share / representation share
= +15.7/ +12.7 (2Wiki), +3.5/ +6.3 (HotpotQA). The control alone already beats the published graph baselines on HotpotQA (83.9 vs 80.4).

Generalization with a frozen configuration

HotpotQA ($1,000 \times 9,811$), zero tuning:

	R@5	FC@5
BM25 (ours)	72.8	49.5
dense te3-small (pub.)	81.3	64.9
HippoRAG 2 (published)	87.1	75.5
CatRAG (published)	89.5	80.4
CATRAG-REIF v4 (4o-mini)	94.7	90.2
CATRAG-REIF v4 (deepseek)	96.0	93.0

- +6.5 R@5 / +12.6 FC@5 over the best published graph system, with every dial, prompt and exemplar frozen from 2Wiki.
- Same index recipe: 87,629 assertions, 48,856 entries, 146k-node graph, ~\$4.7 once.
- Error profile unchanged in kind: 96/98 residual failures are bridges; the missing gold sits at ranks 6–20 or beyond.

Why it works: each principle removes one dependency on depth

1. **Identity in the index** (canonical memory): islands do not multiply with hops; thresholded synonymy does.
2. **Facts as first-class nodes**: mass travels *through* the approved fact at full weight; qualifiers travel inside the unit.
3. **Resources that scale with the chain** (plan $\rightarrow$ repair $\rightarrow$ packing): budget per slot, not per query — hence the inverted gradient on double chains.

The closed form makes the division of labor exact: the walk diffuses; *seeding* chains. The planner decides the seeds; the representation makes every approved fact and discovered entity *reachable* (own passage) and *preferable* ($\mu = 1$).

Limitations and ongoing work

- **End-to-end answering** (EM/F1, Joint Success Rate) pending; published JSR references: 53.0/55.0.
- **MuSiQue** (labeled 2/3/4 hops): direct test of the multiplicative model and of multi-round repair; requires own-passage as a *set* (647 repeated titles).
- Inference stratum ($n=108$) not individually significant; single encoder untested for sensitivity.
- Analyst prompt semantics (slots, $[X]$) defined partly by demonstration; a specification-first prompt is registered as an evolution for smaller models.
- Clinical instantiation (the motivating deployment): unfilled slots $\rightarrow$ grounded abstention; packed sets $\rightarrow$ auditable evidence dossiers.

Conclusions

- Chain-level failure has **structural causes** — unresolved identity and fixed per-query budgets — and both are removable by design.
- A retriever that resolves identity in the index, reifies facts into modifiable nodes, and plans evidence with a bounded three-role LLM pipeline recovers complete chains: FC@5 **97.7** vs 65.8/67.6 (2Wiki) and **93.0** vs 80.4 (HotpotQA, frozen).
- The gains are **attributed** (ablations; no-graph control: representation +12.7/+6.3), **explained** (closed form: seeding chains, the walk diffuses) and **cheap** (~\$0.001/query; open 20B planner nearly ties gpt-4o).
- Everything is reproducible at zero cost: code, prompts, per-question outputs and caches at github.com/bjur27r/hiv_rag (eval-wiki2).